

Pore-Scale Analysis of Residual Oil in a Reservoir Sandstone and Its Dependence on Water Flood Salinity, Oil Composition, and Local Mineralogy

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Supporting Information

ABSTRACT: Core-flooding of clay-containing reservoir sandstones can yield substantial tertiary recovery by reducing the flood brine salinity, associated with a shift toward water-wetting. Spontaneous imbibition experiments, in which this salinity-induced shift is the main driver for additional recovery, can provide insight into the extent and source of the wettability change, especially when combined with pore-scale imaging of changes in residual oil configurations using micro-CT. Spontaneous imbibition experiments were performed on two sister mini-plugs of a reservoir sandstone. To explore the influence of crude oil composition, each mini-plug used a different oil, of similar density and viscosity and of low asphaltene content, but primarily distinguished by their total acid number (TAN) of 3 and <0.1 mg KOH/g oil. Mini-plugs were imaged by micro-CT after each experiment. Additional insight into the origins of the apparent shift to more water-wetting were obtained by preparing polished embedded sections of the mini-plugs for higher resolution SEM imaging and SEM-EDS mineral mapping. These 2D images were registered into the corresponding cross-section of the 3D tomograms of each mini-plug. In this way the salinity-induced release of oil could be overlain and directly compared to the mineral originally contacting it at each location. Results show that the low TAN oil exhibited much less tertiary recovery compared to the high TAN oil. The local mineralogical analysis revealed that oil removal by low salinity brine was less favored from kaolinite than from silicate grains (quartz, K-feldspar, and Na-plagioclase). Nanoscale imaging of grain surfaces also showed that asphaltene films from the high TAN oil were less prevalent on the rock surfaces than for the low TAN oil.

■ INTRODUCTION

Crude oil composition dictates oil viscosity and influences oil–brine interfacial tension and wettability alteration of rock, all of which have a direct and fairly well understood impact on water flood oil recovery via the mobility ratio, capillary number, and contact angle. Enhanced oil recovery by low salinity brine injection acts via in situ changes in crude oil/brine/rock interactions.^{1,2} Accordingly, the links between oil composition and the low salinity effect are more subtle and less well understood. Some researchers have pointed to the importance of the relation between the oil's asphaltene/resin ratio and the content of water-in-oil dispersion in low salinity brine.³ Others have studied the effect of oil composition on low salinity water flooding of core plugs,^{4–14} some of whom have proposed that the oil's acids migrate to the interface and cause recovery by weakening the oil film.^{12–14} Oil component retention on rock surfaces is also responsive to the concentration of acids in crude oil and depends largely on the ratio of total acid and base number, TAN and TBN.¹⁵ Adsorption of organic acids on clays can play a role during low salinity water flooding; components such as amino acids can exhibit greater absorption to clay surfaces in salt-free aqueous solutions than at high salinity.^{16,17}

Mineralogy also has always been recognized as an important factor in determining whether and to what extent the low salinity effect is operative in a given sandstone.¹⁸ Clay minerals—especially but not exclusively kaolinite—were identified as a

necessary component.^{19–22} Plagioclase has been postulated as a grain mineral of possible relevance to low salinity EOR, as it can alter brine pH depending upon salt concentrations.²³ Dissolution of cement minerals has also been discussed as a contributing factor.^{24,25} However, no comprehensive study has been performed to directly compare local mineralogy to local oil release by low salinity brine.

Nondestructive 3D imaging techniques such as micro-CT scanning can provide further pore-scale insight into the effect of oil composition and properties on low salinity flooding. In this paper, micro-CT imaging and analysis were utilized to study mini-plugs of a clay-rich reservoir sandstone after spontaneous imbibition of high and low salinity brine. Sister mini-plugs of 5 mm diameter were restored using two crude oils, primarily distinguished by their high and low TAN. Each mini-plug was micro-CT imaged in its states after spontaneous imbibition of seawater in secondary mode followed by low salinity brine in tertiary mode. The segmented tomograms showed that all mini-plugs possessed very high residual oil saturation after immersion in seawater, implying a substantially oil-wet state. Subsequent immersion in the low salinity brine resulted in a significant tertiary recovery of 5.1%PV for the high TAN oil, but much

Received: July 11, 2017

Revised: October 24, 2017

Published: October 31, 2017



smaller for the low TAN oil. The tomograms were further analyzed to quantify the pore-scale changes in oil saturation, connectivity, interfacial areas, and mean curvature giving rise to these differing responses. Microscopy studies of mini-plugs after spontaneous imbibition experiments provided further insight into the low salinity oil recovery mechanism local to pore wall minerals.

MATERIALS AND METHODS

All mini-plugs in the study were taken from the same piece of friable, clay-rich, North Sea reservoir sandstone, with porosity of 23.8% and absolute permeability of 640 mD (see Supporting Information for mineral composition from XRD). The received rock piece was cleaned of its bulk oil, brine, and surface-bound asphaltenes using toluene, methanol, and azeotropic chloroform/methanol, and then dried. Coring of three sister mini-plugs of 5 mm diameter and 16 mm length (referred to as A, B, and C) was subsequently performed. The two samples of crude oil used were from neighboring fields. The low TAN oil possessed a density of 0.846 g/cm³ and C₇ asphaltene content of 1.1%, with TAN < 0.1 mg KOH/g oil, while the high TAN oil possessed a density of 0.887 g/cm³ and C₇ asphaltene content of 0.43%, with TAN of 3 mg KOH/g oil. Both oils were preconditioned over formation water at 91 °C for 3 days prior to use. The composition of the formation brine (FW), high (HS) and low (LS) salinity brines, and their corresponding contrast brines (containing 0.4 M NaI) are listed in Table 1.

Table 1. Composition of Brines Used in the Experiments

salt	FW (g/L)	HS (g/L)	HS contrast (g/L)	LS (g/L)	LS contrast (g/L)
NaCl	58.44	23.73	0.35	0.5	
CaCl ₂ ·2H ₂ O	5.88	1.50	1.50		
MgCl ₂ ·6H ₂ O	2.03	10.70	10.70		
KCl	1.49	0.76	0.76		
Na ₂ SO ₄		3.98	3.98		
NaHCO ₃		0.67	0.67		
SrCl ₂ ·6H ₂ O	0.53	0.02	0.02		
BaCl ₂ ·2H ₂ O	0.98				
NaI			59.96		59.96
pH	6.1	8.1	7.6	6.4	6.3

Experimental Methods. Each mini-plug was first micro-CT scanned in its cleaned dry state. All scans were performed using HeliScan micro-CT in order to continuously image the majority of the mini-plug height at a voxel size of 2.3 μm. The mini-plug was then saturated with formation brine FW, by vacuum infiltration followed by isostatic pressurization up to 55 MPa for 24 h to dissolve and diffuse out any remaining air. Primary drainage to irreducible saturation S_{wi} was then performed by centrifugation in the high TAN oil for mini-plug A and the low TAN oil for mini-plugs B and C, in all cases at a spin speed of 3750 rpm maintained for 1 h.

Each mini-plug was then aged in a sealed glass vessel in an oven at 91 °C for 11 days. The vessels used for mini-plugs A and B were glass vials filled with their respective crude oil (pre-equilibrated with FW at this aging temperature) and sealed with epoxy. The aging vessel for mini-plug C was of a different design as shown in Figure 1, comprising a test tube with ground glass stopper, containing bulk FW below the (low TAN) oil and mini-plug. The presence of bulk FW reduced the risk in the original design that water from the crude oil escapes through the vial seal during aging, which would lead to progressive dehydration of the connate brine in the mini-plug to rehydrate the oil. Given the very small pore volume, and low S_{wi} of the mini-plug, a slight reduction in water concentration within the comparatively large volume of the surrounding oil could lead to significant decrease in S_{wi} by drying during aging at this relatively high temperature.

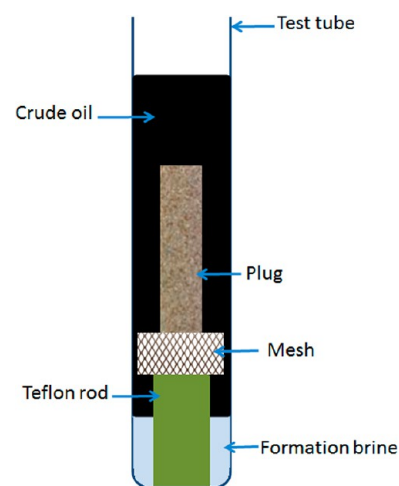


Figure 1. Diagram of aging vessel design used for mini-plug C.

After aging, each mini-plug was immersed in preheated HS brine, which was then maintained at 91 °C for 7 days. This high-salinity spontaneous imbibition step, referred to as SIH, was followed by cooling to room temperature (22 °C) and transfer of the mini-plug into HS contrast brine for 3 days of diffusional exchange of Cl[−] for the dopant ion I[−]. The mini-plug was the rescanned in this contrasted SIH state, mounted in its aluminum holder filled with this same brine. After scanning, the mini-plug was reimmersed in HS brine at room temperature for 2–3 days to re-exchange I[−] for Cl[−], and then reheated to 91 °C and immersed in the preheated LS brine, which was subsequently maintained at this temperature for 7 days. The low-salinity spontaneous imbibition step, referred to as SIL, was then followed by exchange for the LS contrast brine and rescanning in this contrasted SIL state by replicating the above-mentioned procedure.

Each mini-plug was then cleaned using decalin and methanol to remove bulk oil and brine, and rescanned in this dry state after these procedures. It was then resaturated with the HS contrast brine for its final scan in this saturated state. Mini-plug C broke during the centrifugation in decalin used to displace the oil, owing to the fragility of these small friable sandstone samples. Mini-plugs A and B were then soaked in methanol to remove the HS contrast brine, prior to drying and sectioning for FESEM and mineral mapping analysis. Each of these two mini-plugs was then cut crosswise using a Struers saw with a water-based cooling liquid. The smaller piece was used for field emission scanning electron microscopy (FE-SEM) of mineral surfaces near the raw cut or broken surface at very high resolution. The larger of the two cut pieces of each mini-plug was used for mineral mapping using SEM (Quanta 650 FEG) with energy dispersive X-ray spectroscopy (SEM-EDS), which requires a flat, smooth surface for quantitative analysis. For this purpose, each mini-plug piece was embedded in Epofix epoxy and then cut crosswise, followed by polishing and carbon coating. The flowchart in Figure 2 summarizes the experimental workflow.

Image Processing and Analysis. The procedure used for processing and analysis of the series of tomograms (in the states: dry, SIH, SIL, dry postflooding, and brine saturated) comprised six steps. Here “macro-” and “micro-” refer to features (e.g., pore or oil within it) which were resolved at the given voxel size or were subresolution, respectively. The steps were as follows: (1) cylindrical masking, (2) spatial registration to the dry (postflooding) tomogram, (3) two-phase segmentation of macro-oil in the SIH and SIL states, (4) three-phase segmentation of macro-pores, macro-grains, and microporous regions (referred to as clay aggregates) in the dry state, (5) microporosity quantification within the clay aggregate phase using the difference tomogram of brine-saturated minus dry states, and (6) micro-oil and microbrine segmentation within the clay aggregate phase using the difference tomogram of brine-saturated minus SIH or SIL states. Movement of macro-oil and -brine from high to low salinity was analyzed by using the difference tomogram of SIH minus SIL states to segment the four categories of occupancy of voxels within resolved

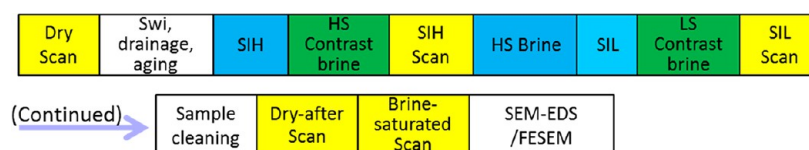


Figure 2. Flowchart of the experimental workflow for each mini-plug.

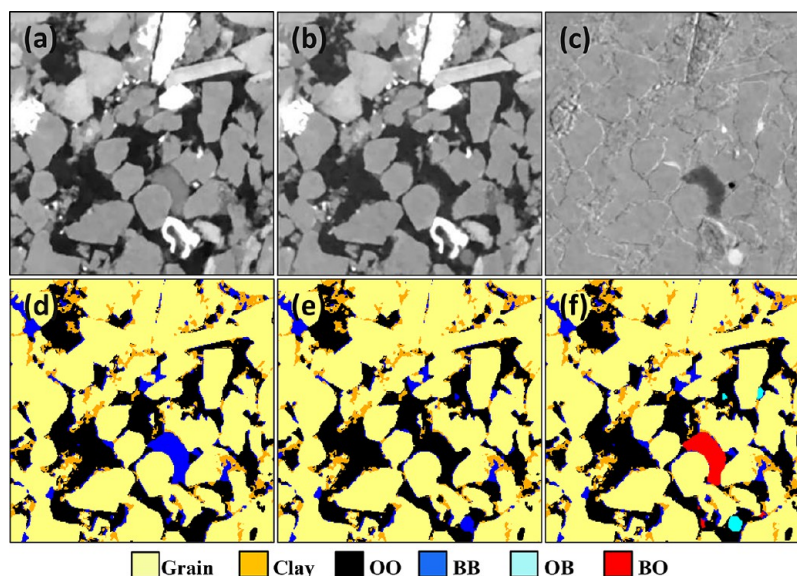


Figure 3. Subarea ($0.63 \text{ mm} \times 0.63 \text{ mm}$) of tomogram of mini-plug C, registered between the states (a) SIH and (b) SIL, also showing (c) their difference, (d) segmentation of SIH, and (e) SIL into macro-oil (black), macro-brine (blue), grain and clay, and (f) after further segmentation of the macro-liquids into their four occupancy classes OO, BB, OB, and BO.

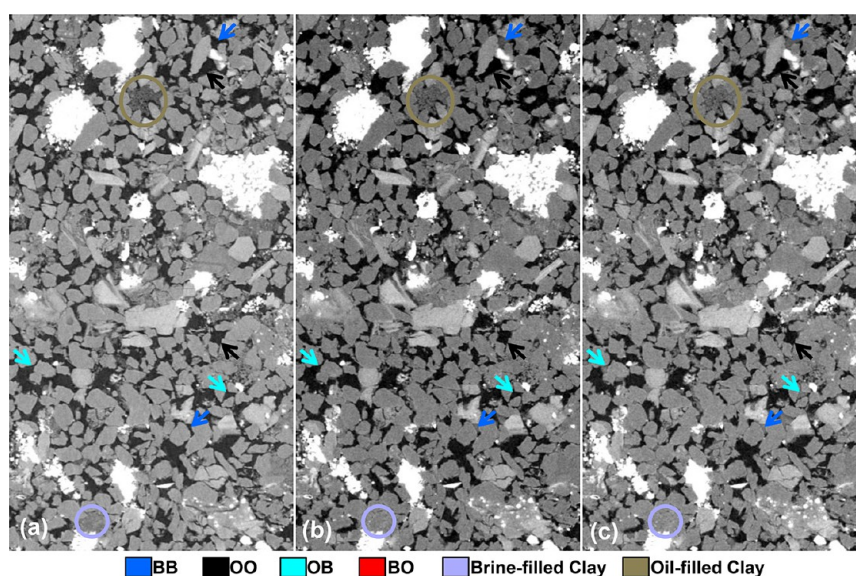


Figure 4. Subarea ($1.6 \text{ mm} \times 3.0 \text{ mm}$) of a longitudinal slice from the tomograms of mini-plug A, registered between the (a) dry state and after (b) SIH and (c) SIL, with examples of the segmented oil and brine phases.

pores, namely, BB (brine-filled in SIH and SIL), OO (oil-filled in SIH and SIL), OB (oil-filled in SIH but brine-filled in SIL), and BO (brine-filled in SIH but oil-filled in SIL).

RESULTS AND DISCUSSION

Figure 3 shows a subarea of a cross-sectional slice of the tomograms of mini-plug C registered in its states after SIH and SIL, and their corresponding four-phase segmentations into

resolved oil, brine, grain, and clay aggregate. They also show the further segmentation of resolved liquid that moved between these two states (OB and BO), obtained from their difference tomogram.

Observations from the Flooded-State Tomograms.

Figures 4 and 5 display a subarea of a central longitudinal slice of the sequence of registered tomograms in the dry, SIH, and SIL states for mini-plug A (with high TAN oil) and B and C (both

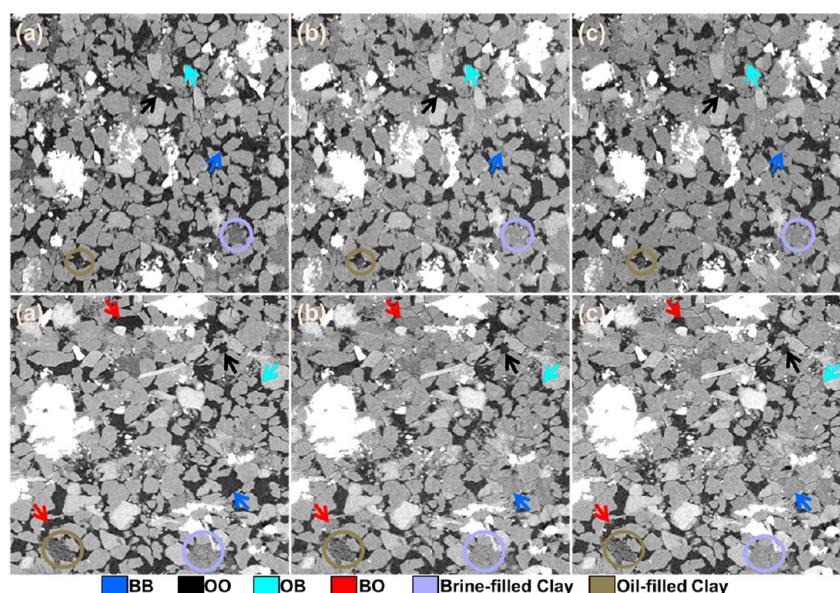


Figure 5. Subarea (1.6 mm $\times$ 1.6 mm) of a longitudinal slice from the tomograms of mini-plugs B (top) and C (bottom), registered between the (a) dry state and after (b) SIH and (c) SIL, with examples of the segmented oil and brine phases.

with low TAN oil). From XRD, the rock comprises (by weight) 33% quartz, 16% feldspar, 7% plagioclase, 27% kaolinite, 10% muscovite/Illite, and 5% siderite. The darkest phase in the dry-state images (a) in Figures 4 and 5 is macro-pore, while the next brighter phase (dark gray) corresponds to pore-filling clays, mainly kaolinite, and the microporosity within them. Quartz and plagioclase grains are brighter, since they do not possess internal porosity; however, these two mineral classes are almost indistinguishable from their X-ray attenuation grayscale. Feldspar (potassium) grains are in turn somewhat brighter, and display some weathering, while muscovite/Illite flakes and aggregates are brighter still, while the brightest grains are siderite.

Features that brighten from the dry to SIH state in images (b) contain contrast brine, since oil and air have similarly low attenuation. Brine primarily occurs in the form of (subresolution) microbrine, which most likely occupied these microporous regions throughout the restoration process. Some examples of (resolved) macro-brine exist (e.g., blue arrows), most commonly in resolved pore throats, and are suggestive of brine advance during SIH from tighter locations originally occupied by connate brine. Based on visual inspection, brine-filled clay aggregates (e.g., purple circles) are generally less porous and tighter than their oil-filled counterparts (e.g., olive circles). Further brightening from the SIH to SIL state in images (c) corresponds to tertiary recovery (OB, e.g., cyan arrows), and is usually manifested as further advance of pore-throat menisci from SIH into pore bodies in SIL. No discernible changes in saturation within microporous regions occur in SIL, so the subsequent quantitative analysis focuses on macro-pores. The vast majority of macro-pore occupancy corresponds to OO (e.g., black arrows), i.e., pores which remain oil-filled throughout SIH and SIL.

While the above observations generally apply to mini-plugs A–C in Figures 4 and 5, some differences exist between these three samples. Tertiary oil recovery (OB) is less frequent in mini-plug B than A. Mini-plug C contains most brine after SIH, with some large pore bodies being substantially occupied by macro-brine in this state. This could indicate that the different aging protocol used for this sample led to a somewhat less oil-wet state

than for B. Instances of OB due to further advance of brine from SIH to SIL are quite frequent in mini-plug C, but are counterbalanced by the reverse effect of withdrawal of brine in pore bodies (BO, e.g., red arrows), which was only very rarely seen in A and B. Thus, the OO category of macro-pore occupancy is least dominant in C.

Image Analysis Results. Table 2 lists the volume averages of each segmented phase, beginning from the bulk volume fractions

Table 2. Volume Fractions of Phases Segmented from the Entire Masked Volume of the Series of Registered Tomograms of Mini-Plugs A–C^a

segmented phase	grain (% BV)	microporous clay (%BV)	macro-porosity (% BV)	micro-porosity (% BV)	total porosity (%BV)		
A	70.2	10.1	19.7	4.6	24.3		
B	68.3	12.6	19.2	4.5	23.7		
C	68.1	11.9	19.9	-	-		

Fluid Saturation							
segmented phase	S_o SIH (%PV)	S_o SIL (%PV)	S_o change (%PV)	OO (% PV)	BB (% PV)	OB (% PV)	BO (% PV)
A	78.4	73.3	5.1	73.5	21.4	5.1	0
B	89.0	87.7	1.3	87.1	11.0	1.9	0
C	71.5	71.4	0.1	68.9	25.9	2.6	2.6

^aPorosity is given as percentage of total bulk volume (BV) and fluid saturations as percentages of total pore volume (PV).

of grains (resolved solid mineral), clay aggregates (including their solid and microporous fractions), and resolved (macro-) pores. Microporosity within the clay was quantified from the difference tomogram of brine-saturated minus dry state; it is not provided for mini-plug C since it broke prior to the final workflow step of brine saturation. According to Table 2, calculated macro-porosity of all samples is consistently about 20% and the total porosity of mini-plugs A and B is about 24%.

Oil saturations (S_o) in Table 2 are given as percentages of pore volume (PV), not of the fraction of it originally occupied by oil (OOIP), since the initial brine saturation S_{wi} after aging was not

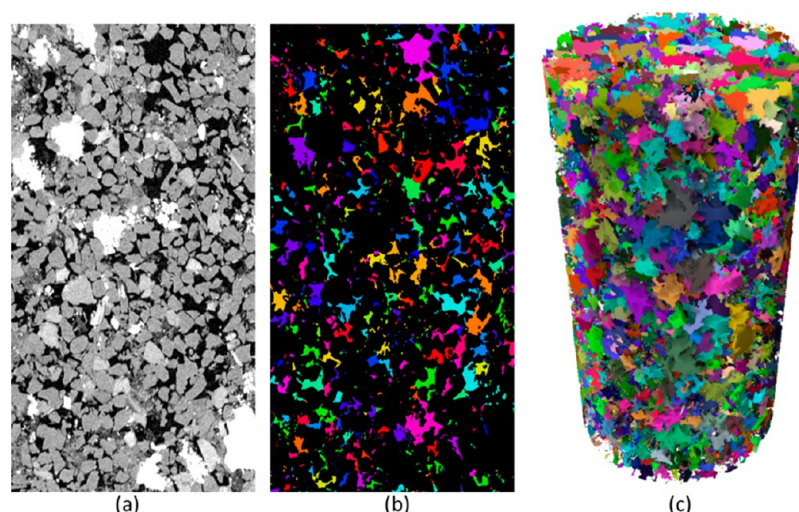


Figure 6. (a) Subarea (1.6 mm $\times$ 3.0 mm) of a longitudinal slice of the dry-state tomogram of mini-plug B, (b) the corresponding pore partitioning in this slice, and (c) a 3D visualization of the partitioning of this same section. Each color represents one single pore body.

measured. All three mini-plugs possess high S_o after SIH compared to the corresponding results after HS water flooding,²⁶ owing to the limited scope for spontaneous imbibition into this substantially oil-wet state. For mini-plug A, its high value of S_{or} after SIH is significantly reduced by 5.1 p.u. after subsequent SIL, with all of this reduction corresponding to OB without any counteracting BO. Oil saturation in mini-plug B after SIH is even higher than for A, and is suggestive of virtually no secondary recovery during SIH. This could be interpreted as being caused by establishment of a more strongly oil-wet state in B, due to the stronger attraction and adhesion of this low TAN oil to the predominantly acidic sites on the silicate minerals, along with its higher asphaltene content compared to the high TAN oil. Subsequent exposure to the LS brine resulted in much less tertiary recovery from B than from A.

To enable a fairer comparison of the role of oil acid number in the low salinity effect for this reservoir sandstone, the experiment for mini-plug B was repeated for its sister C using this same low TAN oil, but with the modification to the aging setup shown in Figure 1, designed to instill a less oil-wet state. This indeed resulted in a lower value of residual oil saturation after SIH than in B, which in fact considerably undershot that in A after SIH. In spite of this, the low salinity effect was again much weaker in C than for the high TAN oil in A. Some oil movement took place from SIH to SIL in C, but the net incremental recovery was close to zero due to the canceling contributions from OB and BO. Thus, irrespective of whether the initial wettability state was more or less oil-wet for the low TAN oil, the low salinity effect was stronger for the high TAN oil.

Pore-Scale Distribution. The resolved pore phase segmented from the dry-state tomogram of the mini-plug was partitioned into its constituent labeled pore bodies using a watershed algorithm,²⁷ as shown in Figure 6. The resolved residual oil phase segmented from the registered tomograms of the SIH and SIL states was overlain onto this partition to calculate the residual macro-oil saturation in each pore body, shown in the scatter plots of Figure 7. In mini-plug A after SIH in Figure 7a, the smallest pores are mainly filled with brine, while somewhat larger pores $\sim 10^{-5}$ mm³ (~ 25 μ m equivalent sphere diameter, ESD) contain more oil than brine, evidencing the oil-wet state. Beyond this size, the distribution bifurcates into a major branch of oil-dominated pores (averaging around 85%

saturation in the largest pores) and a minor branch down to much lower oil saturation, with substantial scatter in between. The most significant change after SIL is that this minor branch strengthens and the scatter increases, resulting in a downward shift in the average oil saturation to around 75% in pores above $\sim 10^{-3}$ mm³ (~ 120 μ m ESD). This is due to the above-mentioned brine advance from small pores in SIH further into larger pore bodies in SIL.

For mini-plug B after SIH in Figure 7b, the transition from brine-filled smaller pores to oil-filled larger ones is steeper than for A, suggesting a more oil-wet state, and the bifurcation at $\sim 10^{-5}$ mm³ pore volume is very one-sided, with most larger pores having very high oil saturation (completely oil-filled to all intents). The only slight change caused by SIL is a small drop in oil saturation in pores $\sim 10^{-4}$ mm³ (~ 60 μ m ESD), as brine advances somewhat from tight pores into corners of these intermediate-sized pores. For mini-plug C after SIH in Figure 7c, more brine is present in small pores (less oil-wet state) and oil saturation is also significantly lower in the largest pores, averaging just over 70%, and with great scatter. After SIL, the distribution changes slightly due to oil displacement from intermediate-sized pores and its reaccumulation in larger pores, in line with the tomograms shown in Figure 5 and the segmentation results of Table 2.

Interfacial Area of Residual Oil. To further understand residual oil changes from SIH to SIL, the contact area of macro-oil and macro-brine with each other and with the external surfaces of the two segmented rock phases of grains and clay aggregates were calculated for the three mini-plugs. The results are given in Figure 8 and the Supporting Information. Figure 8 plots the resolved oil–brine interfacial area normalized by total pore-rock area and its breakdown into contributions from the four occupancy classes. As expected for these systems with high residual oil, its interfacial area to brine increases from SIH to SIL but remains small (usually less than 10%) relative to pore wall area. For mini-plug A, its oil–brine area after SIH is dominated by the fraction (OO-BB) that will remain unmoved after SIL, relative to the smaller fraction (OB-BB) that will move. Oil–brine area roughly doubles from SIH to SIL, as the fraction (OB-OO) of interface created by this oil removal (OB) becomes around the same overall size as the unmoved fraction. Mini-plug B exhibits a similar change, but much slighter, owing to the

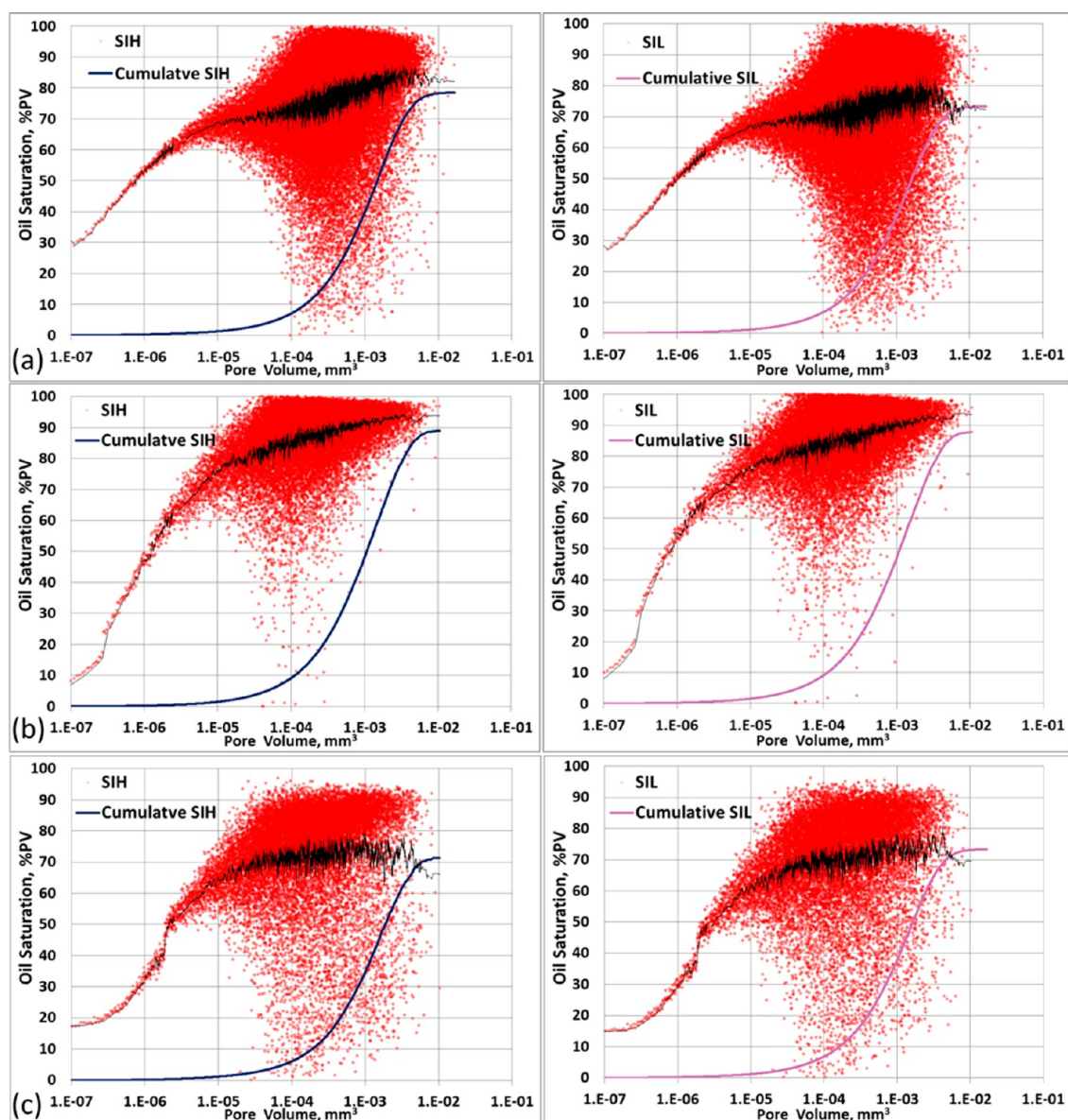


Figure 7. Scatter plot of residual oil saturation in each resolved pore body versus its individual volume (red points), also showing moving average (black line) and volume-weighted cumulative distribution (dark or light purple curve) after SIH (at left) followed by SIL (at right) for the three mini-plugs (a) A, (b) B, and (c) C.

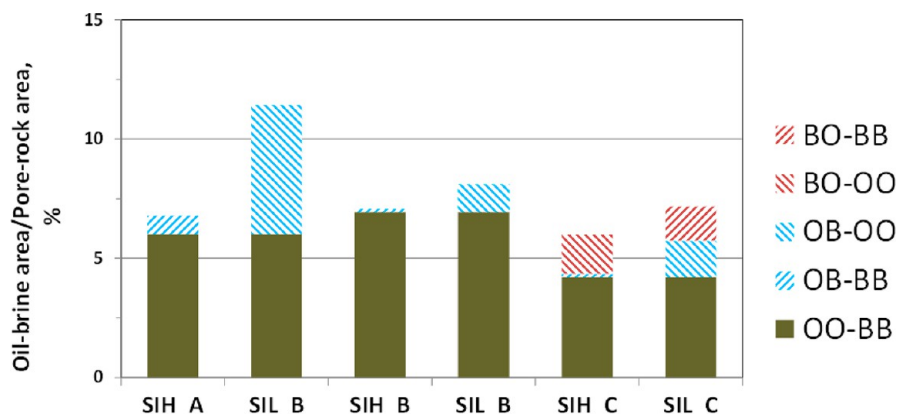


Figure 8. Salinity-induced change in interfacial contact area of macro-oil with brine relative to total pore-rock area.

smaller overall recovery. The results for mini-plug C now also include the non-negligible contribution from oil re-entering

brine-occupied pores (BO). Its unmoved oil–brine interfacial fraction (OO-BB) is much lower, and the fraction that will move

from SIH is dominated by the part (BO-OO) associated with oil re-entry. However, oil–brine interfaces lost from SIH (BO-OO) and created in SIL (BO-BB) by this oil re-entry are of similar sizes, while the fraction lost from SIH (OB-BB) and created in SIL (OB-OO) by oil removal grows substantially and is chiefly responsible for the overall increase in oil–brine area in SIL.

Oil/Brine Interfacial Mean Curvature. To evaluate the shape of the oil–brine interface and to further investigate any shift in wettability from SIH to SIL, oil–brine interfacial mean curvature, H , was calculated using a Monge geometrical algorithm²⁸ after removing the portion of the oil–brine interface lying within 15% of the range of distances to the grain surface. This clipping (Figure 9) served to exclude the transition zone

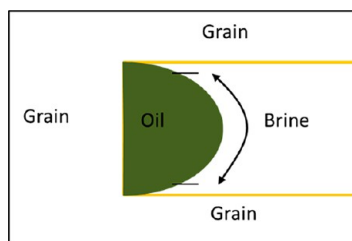


Figure 9. Schematic of central portion of oil–brine interface used in mean curvature calculation.

from oil to grain which often has a high uncertainty in segmentation due to partial volume effects or phase contrast. The resulting average values and standard deviations are listed in Table 3, with the convention that positive or negative mean

Table 3. Average Mean Curvature, H , of the Resolved, Segmented Oil–Brine Interfaces, within the Three Mini-Plugs A–C after SIH and SIL

plug	state	H (voxel ^{−1})	H^{-1} (μm)
A	SIH	−0.032	−73
	SIL	−0.029	−80
B	SIH	−0.028	−79
	SIL	−0.027	−83
C	SIH	−0.071	−32
	SIL	−0.089	−25

curvature corresponds to an interface curving toward or away from oil, respectively. The average value of mean curvature was obtained by fitting to its histogram peak rather than averaging over outputted values from all interfacial patches, in order to exclude the influence of extreme values at both ends. The average mean curvature in Table 3 increases from SIH to SIL for mini-plug A, very slightly increases for B, and decreases for C, consistent with a weakening shift toward water-wet, which is in line with the observed weakening oil recovery by low salinity from A to B to C.

Greater insight into recovery mechanisms from SIH to SIL can be obtained by restricting the analysis to the minority of oil–brine interfaces which moved between these two states. Table 4 gives the results thus derived from separate calculation of mean curvature on each of these four categories of moved interface, or only two for mini-plugs A and B in which instances of oil re-entering brine-occupied pores (BO) were virtually nonexistent. The mean curvature calculation yield similar trends, namely, that interfaces associated with oil removal decrease in curvature from SIH to SIL in all three mini-plugs. In other words, their initial

Table 4. Average Mean Curvature, H , of the Resolved, Segmented Oil–Brine Interfaces of Each of the Four Categories Having Moved from SIH to SIL, within the Three Mini-Plugs A–C

plug	state	interface category	H (voxel ^{−1})	H^{-1} (μm)
A	SIH	OB-BB	−0.021	−111
	SIL	OB-OO	−0.030	−77
B	SIH	OB-BB	0.000	∞
	SIL	OB-OO	−0.031	−72
C	SIH	OB-BB	−0.061	−37
	SIL	OB-OO	−0.077	−29
	SIH	BO-OO	−0.049	−46
	SIL	BO-BB	−0.006	−376

configuration in SIH prior to recovery (OB-BB) is less strongly curved toward water than their final configuration in SIL after recovery (OB-OO). In particular, the average mean curvature of OB-BB menisci (−0.021 voxel^{−1} in Table 4) after SIH in mini-plug A is significantly higher than the overall SIH average (−0.032 in Table 3), which enables low salinity brine to displace oil from these less oil-wet locations. Oil removal in SIL proceeds until the average mean curvature of the corresponding new menisci (OB-OO; −0.030 in Table 4) equilibrates with the overall SIL average (−0.029 in Table 3). The same is true for mini-plug B, in which the average mean curvature of the before-state OB-BB menisci (0.000 voxel^{−1}) lies well above the SIH average value (−0.028), while the corresponding average of the after-state OB-OO menisci (−0.031) lies close to the SIL average value (−0.027). Tertiary recovery from B is thus limited by the closeness of the average mean curvature values in SIH and SIL states, as mentioned above, and also by the relatively small OB-BB area of higher mean curvature over which oil displacement occurs.

The behavior of mini-plug C is distinctly different from A and B; average mean curvature after SIH is much lower and becomes even lower after SIL in Table 3. This is thought to be due to spontaneous formation of stable droplets of water in oil (with strong negative curvature toward water) in SIH and increasingly so in SIL, as can be seen in Figure 3. The creation and swelling of these droplets displaces oil, resulting in reduction in S_o (which is lowest in the SIH state for mini-plug C) or re-entry of oil into brine-occupied pores (which is most common in the SIL state for C). As shown in Table 4, these re-entering BO-BB menisci have higher curvature as the contact angle in this secondary receding mode is lower than for primary advance.

It is not known why these droplets were manifested in mini-plug C but not in B, which also contained this same low TAN crude oil. However, the overall weight of observations points to the high TAN oil in mini-plug A creating a less strongly oil-wet state after aging, making the oil more amenable to release from rock surfaces (especially grains) via wettability shift to more water-wet in the low salinity brine. On the other hand, the low TAN oil in B and C remains more strongly bonded to rock surfaces. In this bonded state, connate water is more likely to be disconnected, in which case its hydration via osmosis during immersion of the mini-plug in brines of lower salinity than the formation water can lead to creation of water droplets. It is also possible that such water droplets are more stable in the low TAN oil. However, this mode of oil displacement by internal water droplet formation is less efficient than meniscus advance via wettability shift as for mini-plug A in the high TAN oil.

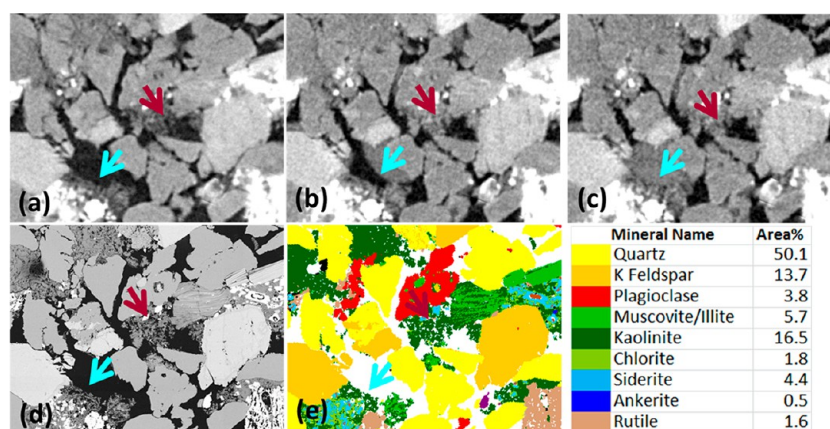


Figure 10. Subarea (0.54 mm $\times$ 0.41 mm) of section of mini-plug A, registered between the (a) dry tomogram, (b) SIH tomogram, (c) SIL tomogram, (d) BSEM image, and (e) SEM-EDS mineral map.

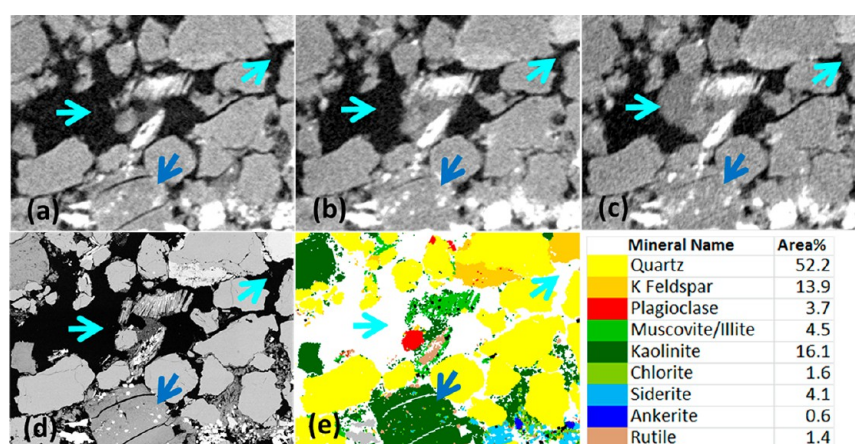


Figure 11. Subarea (0.54 mm $\times$ 0.41 mm) of section of mini-plug B, registered between the (a) dry tomogram, (b) SIH tomogram, (c) SIL tomogram, (d) BSEM image, and (e) SEM-EDS mineral map.

SEM-EDS Mineral Mapping. For each polished section of the 5-mm-diameter mini-plugs A and B, a pair of mosaic images of the entire area were automatically acquired, first with backscattered electrons (BSEM) at 500 nm pixel size using MAPS software, and then with SEM-EDS at 2 μ m pixel size using QEMSCAN software for acquisition and mineral identification. The 2D BSEM image was then registered into its corresponding virtual cross-section through the sequence of 3D registered tomograms of the mini-plug. This then allowed the 2D SEM-EDS mineral map, after its in-plane alignment with the BSEM image, to be registered into the same matching cross-section.

The sequence of registered cross-sectional images allowed the oil distribution after SIH and SIL to be overlain onto the corresponding BSEM image, which serves as a five times higher resolution version of the dry-state tomogram, and most importantly on the SEM-EDS mineral map. In this way, locations of tertiary oil recovery, where pores brighten from the SIH to SIL tomograms due to replacement of oil (dark) by brine (intermediate gray), can be directly compared to the neighboring mineralogy. Two zoomed images from cropped subareas which exhibit tertiary recovery within mini-plugs A and B are shown in Figure 10 and 11. Cyan arrows point to oil removal from tomogram-resolved pores during SIL (category OB), while blue and maroon arrows indicate microporous kaolinite aggregates occupied by brine (brighter than in the dry-state tomogram) or

oil (unchanged grayscale from the dry-state tomogram), respectively.

In the example shown in Figure 10, the fine pores within kaolinite aggregates, whether of high or low porosity, remain brine-filled throughout, although again with oil penetrating the larger pores within looser pore-filling aggregates (e.g., maroon arrow). After SIH, brine has advanced over K-feldspar and quartz grains in one large macro-pore body, and advances further in SIL (cyan arrow), despite the fact that the surfaces of the neighboring kaolinite remain lined with oil.

In the example shown in Figure 11, several kaolinite-rich subregions are identified by SEM-EDS, mainly of low porosity at left and bottom (e.g., blue arrow), with a more porous fine aggregate at center, and looser aggregates of coarser booklets at right, all of which are brine-filled. After SIH, brine has advanced to occupy the pores associated with the agglomerate (at center) of muscovite, kaolinite, and weathered Na-plagioclase and rutile grains, presumably by using their brine-filled fine pores and brine-lined surfaces as a precursor to imbibition. Further advance over quartz and Na-plagioclase grains into the largest pore body occurs during SIL (cyan arrow). Near the top right, a brine droplet formed on a cracked quartz grain after SIH allows subsequent advance during SIL to bridge the pore between this and the neighboring K-feldspar grain (cyan arrow). However, instances of these are relatively infrequent in mini-plug B with its low TAN oil.

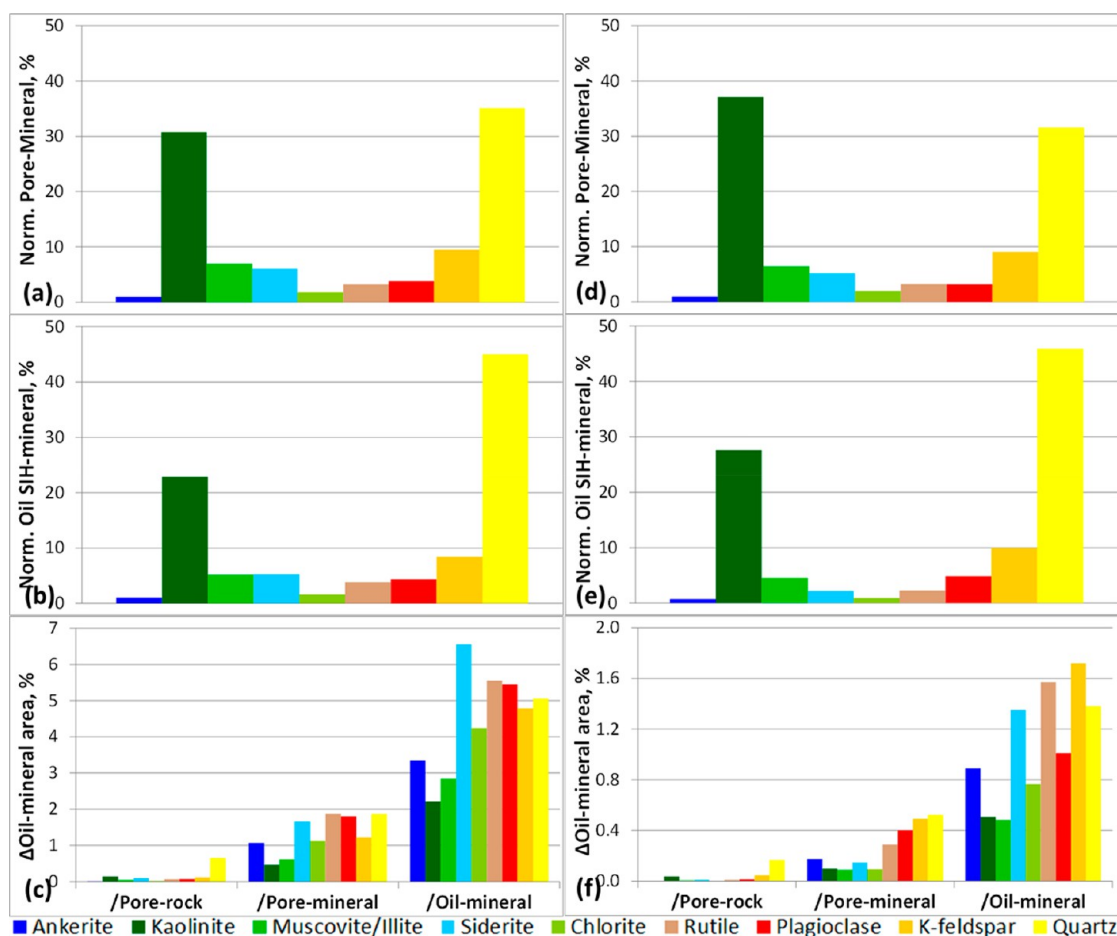


Figure 12. Statistics of pore and oil interfacial area with minerals for mini-plugs A (a,b,c) and B (d,e,f), from overlay of the tomographic oil distributions onto the 2D SEM-EDS mineral maps of sections. (a,d) Pore area of each mineral as percentage of total. (b,e) Interfacial area of each mineral contacting oil after SIH as percentage of total. (c,f) Reduction in interfacial area of oil contacting mineral from SIH to SIL, normalized in three ways.

These overlays not only provide visualizations of oil release from minerals but also allow calculation of oil area per mineral statistics. The quantification was performed by combining the voxel segmentation of the resolved pore and oil phases in the SIH and SIL states with the mineral assignment at each pixel from QEMSCAN to calculate the overall contact area (boundary length) of oil with each mineral over the polished section. Figure 12 presents the results for mini-plugs A (averaged over three polished sections) and B (one section). Graphs (a,d) display the relative surface area of each of the main minerals, i.e., its interfacial area with tomogram-resolved pore space as a percentage of the total area. The surface area of each mineral is strongly correlated to its volume (see Supporting Information), with the exception of kaolinite, which possesses much greater surface/volume at this resolution. As a result, kaolinite and quartz present roughly equal areas, while all other minerals possess substantially less.

Graphs (b,e) in Figure 12 plot the contact area of each mineral with resolved oil in the high salinity state, as a percentage of its total area. Compared to the available pore area in (a,d), proportionally more quartz oil contacts and fewer kaolinite oil contacts in the SIH state of mini-plugs A and B. This is consistent with the above observations that kaolinite is more frequently associated with brine, in the form of connate plus any imbibed brine advancing from it. Of the other minerals, oil is slightly under-represented on the surfaces of muscovite and siderite, while it is slightly over-represented on Na-plagioclase. Graphs

(c,f) plot the reduction in contact area of each mineral with oil from SIH to SIL, normalized in three ways, namely, relative to (i) total interfacial area of all minerals with pore space (i.e., the normalization used in graphs (a,d)) or (ii) interfacial area of the mineral in question with pore space (i.e., the value for the mineral in graphs (a,d)) or (iii) interfacial area of the mineral in question contacting oil after SIH (i.e., the value for the mineral in graphs (b,e)).

According to the first normalization for mini-plug A in Figure 12c in the leftmost set of bars, oil releases from a total of 1.2% of total rock surface area from SIH to SIL. More than half of this low salinity-induced release is from quartz, as expected for this most predominant mineral, which has 4.5 times greater area of release compared to kaolinite (the second most prevalent mineral). After normalizing by the pore area of each mineral, the distribution (in the middle set of bars in Figure 12c) is generally much more uniform, e.g., Na-plagioclase becomes almost on par with quartz, although kaolinite remains 4 times lower than quartz. The rightmost set of bars in Figure 12c provides the fairest assessment of the true tendency for oil release during SIL from each mineral, when taken relative to its oil contact after SIH. Since oil after SIH is over- and under-represented on quartz and kaolinite surfaces, respectively, this final normalization further reduces the count of relative oil release from quartz and increases that from kaolinite. However, kaolinite remains the lowest of all minerals, with less than half the relative oil release of quartz, and the second most prevalent clay, muscovite, is second lowest. Both

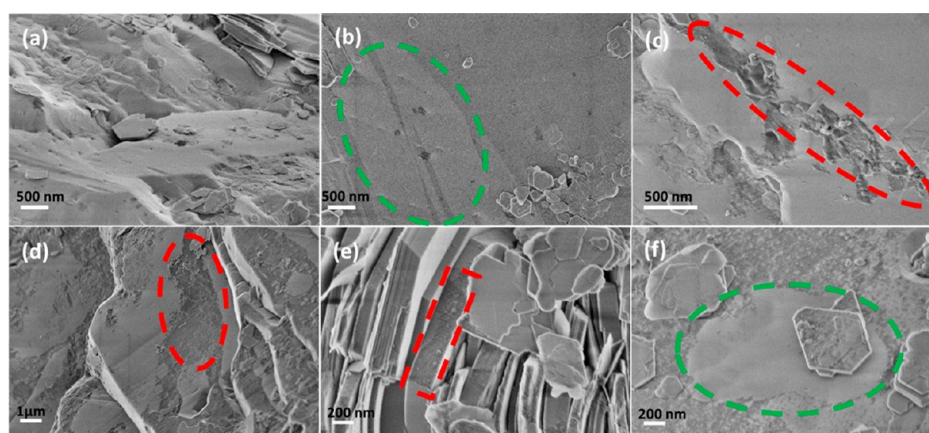


Figure 13. FESEM images of mineral surfaces of mini-plugs A (a–c) and B (d,e) near a raw surface.

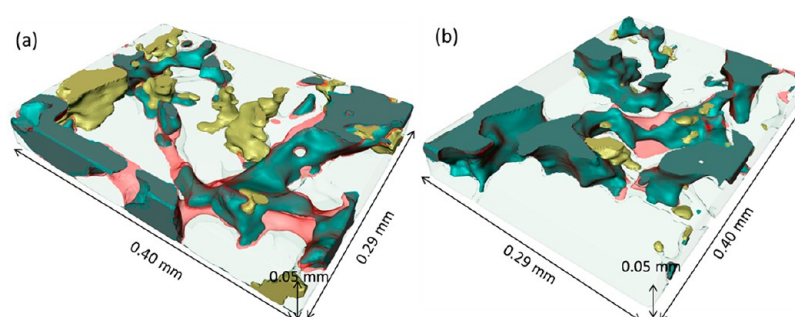


Figure 14. 3D visualizations of SIH and SIL states in subvolumes of mini-plugs (a) A and (b) B. Residual oil after SIH and SIL is red and green, respectively, clays are yellow, and grains are transparent.

of these clays released less than 3% (area-wise) of their originally contacting oil. The low salinity effect from spontaneous imbibition in mini-plug A thus does not appear to be directly related to preferential release from clays. In fact, release is most preferred from surfaces of siderite, followed by rutile and Na-plagioclase.

The corresponding normalizations of the reduction in contact area of each mineral with oil from SIH to SIL for mini-plug B with the low TAN oil are shown in Figure 12f. As expected from the very weak tertiary recovery, the overall area of oil release is much smaller than for mini-plug A. All of the 9 main minerals in mini-plug A released at least 2% of their attached oil by low salinity brine, whereas all in B showed less than 2% of oil removal. However, the three normalized measures per mineral are each reasonably well correlated between A and B, so that the latter behaves roughly as a scaled-down version of the former. In particular, kaolinite and muscovite again show the least removal relative to their interfacial areas with pore space and with oil after SIH. It thus appears that the effect of the oil's TAN on tertiary recovery in low salinity brine does not stem from specific and local unlocking of the intermolecular bonds between oil acid groups and particular mineral surface groups.

FE-SEM Imaging of Pore-Scale Wettability. The raw surface of the sister half of mini-plug A and B was imaged using FE-SEM up to very high resolution to discern the local presence or absence of asphaltene films on mineral surfaces, which dictate the pore-scale wettability and which remain bound to the surfaces after removal of the overlying bulk oil and brine. A selection of such images are shown in Figures 13. Locally water-wet (clean) subareas and more oil-wet subareas bearing adsorbed/deposited asphaltenes are indicated by green and red

dashed lines, respectively. Figure 13a shows a quartz or Na-plagioclase grain surface littered with kaolinite platelets; parts of the grain surfaces at lower right and upper left bear a thin, patchy deposit of asphaltene, while the remainder appears clean. Image (b) is of a euhedral grain surface, the right half of which is covered by a very thin film of asphaltene which also decorates the kaolinite platelets, while the brighter region circled in green appears clean. The grain shown at higher magnification in (c) is largely clean, aside from occasional crusty patches of asphaltene deposit circled in red.

For mini-plug B in Figure 13, around half of the facets of the quartz or Na-plagioclase grain in image (d) bear an uneven, crusty deposit (one part of which is circled in red). Image (e) shows that many edges of kaolinite platelets in this booklet bear an asphaltene film, while the basal planes appear cleaner, due to less chemical affinity for oil and/or stronger capillary forces holding brine in these crevices. While the majority of the grain surface in (f) is covered with asphaltene, clean spots in their midst are clearly apparent and also in (d). The more rounded examples of these clean spots presumably correspond to brine droplets which prevented asphaltene deposition during aging, while more polygonally shaped spots were probably protected from oil exposure by lining kaolinite platelets, which were subsequently stripped from the grain during oil recovery.

The overall impression from the FE-SEM images is that a greater fraction of surfaces in mini-plug B are coated with asphaltene, which is also generally thicker than in A. These qualitative observations of greater oil-wetness of B are in line with the lesser recovery and much weaker low salinity effect seen from micro-CT imaging and analysis. It appears that the greater asphaltene content and low TAN of the oil in B leads to a heavier

deposition of asphaltene during aging, which precludes spontaneous imbibition during SIH and reduces the scope for wettability shift to more water-wet by removal of the deposit during SIL. In comparison, the thinner, patchy deposit from the high TAN oil in mini-plug A is likely to be more conducive to recovery during SIL, at least from the locations where oil is most weakly anchored.

3D Visualizations. To visualize in 3D the changes from SIH to SIL, subvolumes of the segmented tomograms containing instances of oil movement were rendered using Avizo software and shown in Figure 14. Figure 14a shows oil ganglia breaking into smaller blobs during SIL within mini-plug A, primarily due to advancing brine causing release of oil from grain surfaces and snap-off in pore throats between grains, while oil remains attached to the outer surfaces of clay aggregates. Figure 14b shows instances of oil detachment from grain surfaces in mini-plug B, which were far less common than in A.

CONCLUSION

Spontaneous imbibition of high and low salinity brine into a restored, clay-rich reservoir sandstone was studied by tomographic and microscopic imaging and analysis. Two crude oils with differing total acid number (TAN) were used in conjunction with three sister mini-plugs. The high TAN oil exhibited the greatest tertiary recovery by spontaneous imbibition of low salinity brine. The additional recovery was mainly the result of brine advance from pre-existing menisci in throats and small pores to snap off oil in throats and enter larger pore bodies. The salinity-induced wettability shift was sufficient to displace oil from locations that were already more water-wet in the state after spontaneous imbibition of high salinity brine, but was insufficient to cause oil movement from more oil-wet locations. The low TAN oil exhibited much less tertiary recovery, irrespective of whether oil saturation after high salinity exposure was higher or lower than that of the high TAN oil. This low TAN oil (which also possessed more asphaltenes) appeared to induce a more strongly oil-wet state after aging, which largely precluded recovery by brine meniscus advance, but which instead favored oil displacement by the alternative mechanism of water droplet creation and growth in larger oil-filled pores. This droplet formation, thought to be caused by osmosis, appeared to be a much less effective means of recovering additional oil than meniscus advance by wettability change.

SEM and SEM-EDS images of polished sections of the mini-plugs were registered into their corresponding virtual slice through the tomograms to overlay the distribution of residual oil after high and low salinity onto its corresponding mineral map. After spontaneous imbibition of high salinity brine, less oil adhered to kaolinite than to quartz, relative to their available areas. Oil removal by subsequent spontaneous imbibition of low salinity brine was less favored from kaolinite (and from muscovite) than from silicate grains (quartz, K-feldspar, and Na-plagioclase). Release from minerals was naturally much greater for the high TAN oil with greater low salinity effect than for the low TAN oil; however, no particular mineral(s) contributed disproportionately to this difference, suggesting that the dependency of the effect on acid number is not markedly mineral-specific. Complementary ultrahigh resolution FE-SEM imaging of raw cut sections of the mini-plugs confirmed that asphaltene films were less prevalent, thinner, and more irregularly distributed on the rock surfaces after low salinity exposure for the high TAN oil. This suggested a stronger shift toward water-wet, in line with the tomographic evidence.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.energyfuels.7b01978.

Mineral composition of the reservoir sandstone from XRD; modal mineralogy of each cross-section; interfacial area of liquid (oil and brine) contacting grain and clay aggregate (PDF)

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The authors declare no competing financial interest.

ACKNOWLEDGMENTS

Financial support from the member companies of the Digital Core Consortium Wettability Satellite and Statoil are acknowledged.

ABBREVIATIONS

Micro-CT, X-ray microtomography; TAN, total acid number; TBN, total base number; SIH, spontaneous imbibition of high salinity brine; SIL, spontaneous imbibition of low salinity brine; SEM, scanning electron microscopy; BSEM, backscattered scanning electron microscopy; SEM-EDS, energy dispersive X-ray spectroscopy; FE-SEM, field emission scanning electron microscopy; FW, formation water; HS, high salinity; LS, low salinity; BB, brine-filled in SIH and SIL; OO, oil-filled in SIH and SIL; OB, oil-filled in SIH but brine-filled in SIL; BO, brine-filled in SIH but oil-filled in SIL; OOIP, original oil in place; p.u., percentage units; S_{wi} , initial water saturation; S_o , oil saturations

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